



AAKASH SIVAKUMAR

MECHATRONICS GRADUATE

Mechatronics Engineer | Robotics & Software Developer A 2025 Mechatronics Graduate with a specialized focus on Autonomous Systems, Control Systems, and Full-stack Development. Proven experience in implementing ROS2 frameworks, industrial automation, and real-time data management. Adept at bridging the gap between mechanical hardware and software logic, with a growing portfolio in JavaScript-based applications.

Contact



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Github

<https://github.com/ProEvolution03>

Education

2021 - 2025

BTech - Mechatronics **6.63 CGPA**

SASTRA DEEMED UNIVERSITY,
Thanjavur

2019 - 2021

Class 12 - 85.2%

PSBB Senior Secondary School,
K K Nagar, Chennai

2018 - 2019

Class 10 - 83.6%

PSBB Senior Secondary School,
K K Nagar, Chennai

Languages

English

Tamil

Experience

July 2025 - Sep 2025

Mechatronics Engineer Intern - Nirbhav Automation Pvt. Ltd.

- Developed automation solutions for the secure packaging of hazardous chemicals, focusing on green methodology.
- Optimized electronic systems to support high-speed industrial automation processes.

June 2024

Brakes India Private Ltd. - Intern Trainee

- Analyzed the manufacturing and assembly of drum brakes, improving workflow understanding in high-volume manufacturing environment.

Projects

Jan 2026

TaskMaster : Fullstack Task Management Application

- Architected a responsive task management system using React.js for the frontend and Supabase for real-time database management and user authentication.
- Implemented state management and RESTful API principles to ensure seamless data persistence.

Jan 2025 - May 2025

Development of Robotic Lower Limb Model using Simscape

- Designed a robotic lower leg model for hemiparetic patients to simulate and verify the efficacy of walking aids.
- Applied manipulator kinematics and control concepts using MATLAB Simscape.

Sep 2024 - Oct 2024

TurtleBot SLAM Implementation with ROS2

- Integrated Nav2 and SLAM (Simultaneous Localization and Mapping) for autonomous robot navigation.
- Executed precise localization within a mapped environment using Gazebo and Rviz2 simulations

Mar 2024 - Apr 2024

3-DOF Manipulator Design and Build

- Designed and built a 3-DOF manipulator for pick-and-place tasks, gaining hands-on experience with servo motors and Arduino.
- Optimized the manipulator for accurate object handling.

Skills

- **Programming Languages**

Python, C/C++, Java,
Javascript (ES6+), MATLAB.

- **Robotics & Control**

ROS2 (Humble), RoboDK,
Gazebo, Rviz2, SLAM, Nav2,
Simscape, Arduino, PLC, HMI

- **Web Development**

React.js, HTML, CSS

- **Databases & Tools**

React.js, MySQL & MongoDB,
OpenCV, Git/Github. Linux
(Ubuntu),

Courses & Certifications

Oct 2023 - Dec 2023

Robotics Training On-Campus

SASTRA University & Ministry of Heavy Industries of India (MHI)

- Specialized with hands-on experience in AMR's, Scara, COBOT.
- Learnt the methodology on robot chassis assembly

Club Work

Lead - Team Technotainment

The Entertainment and Management Cluster of SASTRA University

- **Flag Wars:** Orchestrated an inter-college event utilizing a live database system to handle 400+ parallel inputs with a dynamic real-time scorecard.
- **Moneyball:** Developed a stock market simulation for 200+ live users, focusing on data synchronization and user interface stability.